

Lab Week 11: Social Engineering

Phishing Emails in Action



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Phishing Emails in Action

Learn the different indicators of phishing attempts by examining actual phishing emails.

30 min

107,758



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Task 1 Introduction

In the first room of the [Phishing Analysis module](#), [Phishing Analysis Fundamentals](#), you learned how to investigate email headers and bodies to identify common red flags. Now, we transition to practice by examining real phishing email samples. This room focuses on the tactics attackers use to mirror legitimate communications. By analyzing real samples, you will learn to identify the subtle nuances that distinguish a routine notification from a sophisticated credential harvesting attempt. Note: The samples throughout this room contain information from actual emails. Proceed with caution if you attempt to interact with any IP, domain, or attachment.

Objectives

- Identify common social engineering tactics used in phishing
- Analyze red flags contained within phishing emails
- Detect link manipulation and tracking pixels
- Deconstruct credential harvesting and attachment manipulation

Prerequisites

- Check out [Phishing Analysis Fundamentals](#) for an overview of email communications and analysis

Answer the questions below

I am ready to analyze phishing emails!



No answer needed

✓ Correct Answer

Task 2 Cancel Your Order

In this task, we will examine a sample email designed to mimic an official transaction receipt from PayPal. By analyzing this specific sample, we will focus on how attackers leverage spoofed email addresses to impersonate trusted services and the strategic use of URL shortening services to obfuscate the final destination of malicious links.

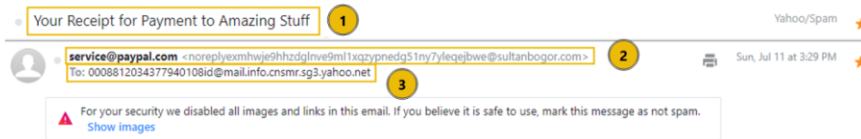
Phishing Techniques Used

- **Spoofed email address:** Mimicking a trusted service to gain immediate credibility
- **URL shortening:** Using redirection services to hide the true destination of a link
- **Branded HTML:** Impersonating legitimate corporate imagery to create a sense of authenticity

First Observations

Let's first look at the sample email header and take a few notes on our immediate observations.

1. **Attention-grabbing subject line:** The subject line uses a fake transaction to create a sense of urgency, prompting you to react in haste
2. **From address:** This is an immediate red flag as the sender details `service@paypal.com` do not match the actual address `gibberish@sultanbogar.com`
3. **To address:** This is an unusual email recipient address and not a normal Yahoo domain



Answer the questions below

Who is listed as the **Merchant** in the email body?

Amazing Stuff

✓ Correct Answer

Task 3 Track Your Package

The next email we will investigate mimics a formal shipping notification to trick recipients into a sense of urgency. By analyzing this sample, we will uncover how attackers use spoofed addresses, link manipulation, and tracking pixels to compromise users. We'll also examine the raw source code to see why email providers flagged these specific red flags.

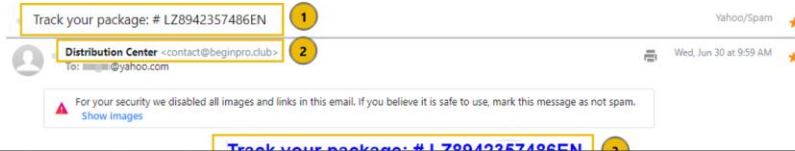
Phishing Techniques Used

- **Spoofed email address:** Mimicking a trusted distribution center to gain immediate credibility
- **Pixel tracking:** Embedding invisible images to notify the sender when the email is opened
- **Link manipulation:** Masking a malicious destination with a fraudulent tracking number

First Observations

Let's take a look at the email below and try to highlight our initial observations.

1. **Subject line:** The subject line uses a fake tracking number to create a sense of urgency, prompting the recipient to click to see their package status
2. **From address:** This is an immediate red flag, as the display name **Distribution Center** does not match the actual sender address `contact@beginpro.club`
3. **Hyperlink:** The link in the email body matches the subject line, although we do not know where it directs to just yet



Answer the questions below

What root domain does the hyperlink in the above example point to?
Be sure to defang the URL.

 Correct Answer

Task 4 Download Document Here

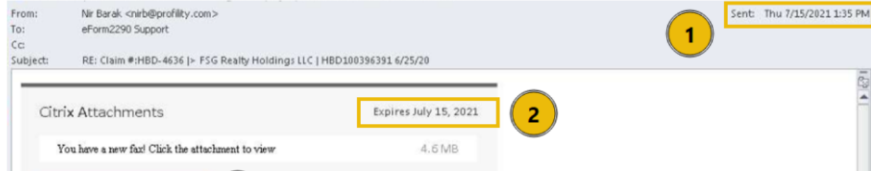
In this task, we will analyze a [phishing campaign](#) that utilizes a multi-stage redirection chain to harvest user credentials. This sample demonstrates how attackers leverage the reputations of professional document-sharing services, such as OneDrive, Adobe, and Microsoft, to create a deceptive path that leads the victim to a fraudulent login portal.

Phishing Techniques Used

- **Artificial urgency:** Creating a narrow window for action to create a sense of urgency
- **Brand impersonation:** Layering trusted brands, like Microsoft and Adobe, to build a false sense of security
- **Link redirection:** Using a chain of URLs to hide the final malicious destination from basic email filters
- **Credential harvesting:** Deploying a fake login portal to capture and exfiltrate usernames and passwords

First Observations

1. **Send date:** The email was sent on Thursday, July 15th, 2021
2. **Expiration date:** A sense of urgency is introduced in this email. Notice that the link to download the fax document expires on the same day
3. **Download Document Here button:** There is an action to perform. In this case, a button to download the fax



Answer the questions below

The attacker deployed a fake portal to capture and exfiltrate user credentials.
What is this type of attack called?

Credential Harvesting

✓ Correct Answer



Task 5 Your Account Is on Hold

In this example, we will again examine an email claiming to be from a trusted, household brand that demands immediate action from the recipient. While previous cases may have relied on malicious links, this variant introduces a shady attachment. By masquerading as an official billing notification, the attacker attempts to bypass standard email filters and leverage the victim's sense of urgency.

Phishing Techniques Used

- **Spoofed email address:** The sender's display name is set to **Netltx billing** to appear legitimate
- **Sense of urgency:** Using a suspended account notification to pressure the victim into acting quickly
- **Brand impersonation:** Utilizing HTML templates and logos to mimic Netflix billing
- **Poor grammar and typos:** Noticeable misspellings of Netflix
- **Attachments:** Using a file attachment rather than a direct link to hide the malicious URL

First Observations

1. **Email subject:** The receiver's ID was suspended and must act quickly
2. **From address:** The display name does not match the user and domain
3. **Brand impersonation:** The email utilizes rendered HTML to impersonate Netflix



Answer the questions below

What is the actual sender email address hidden behind the **Netltx billing** display name?

 Correct Answer

Task 6 Your Recent Purchase

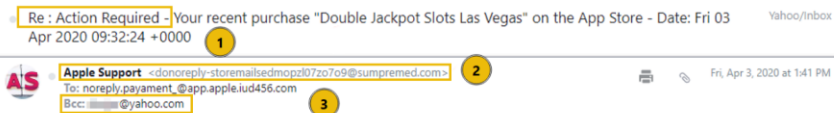
In this example, we will analyze a phishing attempt that masquerades as a billing notification from a major service provider. Unlike previous tasks that relied on body text to lure the victim, this sample utilizes a completely blank email body, and relies solely on a suspicious attachment. We will examine how attackers use BCC fields and spoofed domains to hide the true recipient list while leveraging a sense of urgency to trick the user into opening a malicious file.

Phishing Techniques Used

- **Spoofed email address:** The sender's display name is set to **Apple Support**
- **Recipient is BCCed:** The victim is not directly sent the email
- **Urgency:** Relies on the use of **Action Required** and a fake purchase notification
- **Poor grammar and typos:** Noticeable spelling errors within the email header
- **Attachments:** The email contains a **.dot** file (Microsoft Word Template), which is an unusual format for a receipt

First Observations

1. **Email subject:** The recipient is told they must act quickly to resolve an unauthorized purchase, creating a false sense of urgency
2. **From address:** The display name, **Apple Support**, does not match the user and domain. There are also typos in the From and To addresses
3. **Blind Carbon Copy:** The recipient was not directly emailed, instead **Blind Carbon Copied** (BCC)



Answer the questions below

What does the acronym BCC stand for?

✓ Correct Answer

What is the file extension of the attachment?

✓ Correct Answer

Task 7 Scheduled Shipment

In this example, we analyze a phishing attempt that masquerades as a global shipping notification using spoofed addresses and branded HTML to appear legitimate. We will examine how the attacker funnels the victim toward a malicious Excel attachment designed to execute a hidden payload.

Phishing Techniques Used

- **Spoofed email address:** The sender's display name is set to **DHL Express**
- **Brand impersonation:** Utilizing HTML templates and logos to mimic DHL
- **Attachments:** An Excel document that triggers executable code upon opening

First Observations

1. **Email subject:** Gives the impression that DHL will be shipping a package
2. **From address:** The display name, **DHL Express**, does not match the user and domain
3. **Brand impersonation:** The HTML in the email body is designed to look like it is sent from DHL



Answer the questions below

What is the name of the executable that the Excel attachment attempts to run?

regasms.exe

✓ Correct Answer

Task 8 Conclusion

In this room, you analyzed a series of [phishing](#) attempts and developed the skills needed to distinguish legitimate communications from malicious ones. By examining email headers, body content, links, and attachments, you learned how to identify common red flags such as spoofed senders, suspicious domains, and deceptive [social engineering](#) techniques.

Further Learning

This is the second room of the [Phishing Analysis](#) module of the [SOC Level 1 Path](#). You are encouraged to continue your learning with the walkthrough and challenge rooms below:

- [Phishing Analysis Tools](#)
- [Phishing Prevention](#)
- [The Greenholt Phish](#)
- [Snapped Phish-ing Line](#)

Answer the questions below

Complete the room and continue on your cyber learning journey!

No answer needed

✓ Correct Answer